

You are evaluating ontology reuse and namespace ownership. This is a non-interactive task. Use the GPT 5.5 High Model. Return exactly one valid downloadable JSON file and nothing else.

Do not:

- ask questions or request clarification;
- output Markdown, comments, or code fences;
- explain your reasoning outside the required JSON fields;
- summarize the ontology outside the schema.

Task

Analyze how FASTO reuses external ontologies and namespaces. Identify:

- all reused ontologies and external ontology namespaces supported by the evidence;
- the import or reuse mechanism used for each ontology;
- whether reused content is imported, copied, partially reused, indirectly imported, or reused through another mechanism;
- entities originating from reused ontologies;
- entities apparently created by FASTO inside a reused ontology namespace;
- reused entities that FASTO redeclares, annotates, constrains, or otherwise modifies;
- entities whose namespace ownership is ambiguous, suspicious, or inconsistent;
- reuse, provenance, import, namespace, versioning, and maintainability issues.

Pay particular attention to FASTO-specific classes, properties, individuals, or axioms that use a reused ontology namespace instead of FASTO's namespace.

Evidence

Five evidence sources are provided (24 files).

1. Primary evidence: The original FASTO_RDF.owl in RDF/XML syntax. This is the authoritative source.
2. Secondary evidence: A rule-based verbalizer generated from a temporary sanitized copy of FASTO. This exists only to improve readability.
3. Supplementary evidence: The paper describing FASTO. Use the paper to clarify:
 - which ontologies FASTO claims to reuse;
 - which entities or concepts are described as FASTO-specific;
 - whether the authors describe importing, adapting, copying, extending, or partially reusing external ontologies;
 - intended provenance or ownership.
4. Additional .ttl and .owl files of the reused ontologies: FHIR, DMTO, OntoFood and SSN.
5. Supplementary evidences: .csv, .txt, and .tsv, files, which are the results of standard tools (e.g., ROBOT) and manual investigation (e.g., RDFlib). Treat these files as candidate and baseline observations about ontology reuse, imports, namespace

ownership, and entity provenance. Use them to guide and extend the analysis, but do not reproduce their conclusions without evaluating them against the primary RDF/XML and the FASTO paper.

Evidence rules

- RDF/XML is always the primary source.
- Enhance the results of the additional supplementary sources (results of standard tools and manual inspection), if those are not sufficient.
- The verbalizer is supplementary evidence only.
- Do not compare the RDF/XML and verbalizer.
- The paper must not override explicit RDF/XML evidence.
- If the paper and RDF/XML are inconsistent, report the inconsistency and treat the RDF/XML as authoritative for what is actually encoded.
- Use the additional supplementary evidence to guide, corroborate, or extend the analysis.
- Do not simply reproduce findings from ROBOT, RDFLib, or other supplementary files.
- Verify supplementary observations against the RDF/XML and paper whenever the relevant evidence is available.
- Report only conclusions supported by the supplied evidence.
- Do not invent source-ontology contents, ownership, versions, import histories, or intended semantics.
- The use of an external namespace alone does not prove direct copying or local creation.
- The absence of owl:imports alone does not prove direct copying.
- Distinguish directly reused, copied, locally created, and locally modified entities where the evidence permits.
- Use exact IRIs, namespaces, declarations, import statements, axioms, or paper statements as evidence.
- Use lower confidence and an unclear classification when ownership, reuse scope, or import mechanism cannot be determined reliably.
- Do not force a definitive classification from incomplete evidence.

Entity inventory rules

The classes, object_properties, data_properties, and individuals arrays under directly_reused_entities, local_entities_in_reused_namespace, and entities_with_unexpected_namespace are exhaustive entity inventories.

For every such array:

- Include every qualifying entity supported by the evidence.
- Each array element must be exactly one complete entity IRI.
- Do not combine multiple IRIs in one element.
- Do not include counts, descriptions, evidence, source locations, qualifications, or explanatory prose.

- Do not replace complete inventories with representative examples.
- Do not use phrases such as "including", "such as", "among others", "related entities", "entity set", or "N entities".
- Do not use ellipses, "etc.", placeholders, or omission statements.
- Preserve each IRI exactly as encoded.
- Remove exact duplicates.
- The same entity may appear in more than one category when multiple category definitions apply.
- Do not merge BFO, DDO, and DMT0/FDTM entities together.

Axiom array rules

- Use one distinct axiom or one precisely defined homogeneous axiom pattern per element.
- Identify the affected entity IRI, axiom type or predicate, and object or restriction when available.
- Do not replace identified axioms with counts and selected examples.
- Group an axiom pattern only when the same reasoning applies and all affected entities are already individually enumerated in the applicable entity arrays.

Issue grouping rules

The exhaustive-enumeration requirement does not require one detected issue per entity.

Entities may be grouped only inside detected_issues, and only when they share the same:

- issue type;
- namespace role;
- provenance assessment;
- modification status;
- severity;
- evidential reasoning;
- explanation;
- suggested fix;
- confidence level.

For a grouped issue:

- describe the entity set precisely in the entity field;
- do not repeat the complete IRI inventory in the issue;
- ensure every affected entity is individually listed in the applicable entity arrays.

Create separate issues when entities differ materially in provenance, modification status, entity type, namespace role, issue type, severity, evidence, explanation, suggested fix, or confidence.

Reporting rules Report entity-level issues separately when entities differ in:

- provenance;

- modification status;
- namespace role;
- issue type;
- supporting evidence.

Entities may be grouped only inside detected_issues when the issue-grouping rules are satisfied. Entities must never be grouped, counted, summarized, or represented by examples in the entity inventory arrays. Do not merge findings merely because entities share a namespace. Remove exact or substantively equivalent duplicate issues.

Output rules

- Return exactly one JSON object conforming to the schema.
- Do not create fields outside the schema.
- Include all required objects and arrays, even when empty.
- Escape all evidence as valid JSON strings.
- Use confidence values from 0.0 to 1.0.
- Use globally unique sequential issue IDs: RNS1, RNS2, RNS3, and so on.
- Do not restart issue numbering for each reused ontology.
- When expected_original_namespace cannot be established from the evidence, use an empty string and explain the uncertainty in the relevant assessment.

Field rules

import_method: This field is an array. Include one or more of:

- complete_import
- partial_import
- indirect_import
- direct_copy
- no_import_detected
- unclear

Use multiple values only when the evidence supports multiple reuse mechanisms for the same ontology. reuse_scope: Select exactly one:

- complete
- partial
- unclear

directly_reused_entities: List entities or axioms supported as originating from or being directly reused from an external ontology.

local_entities_in_reused_namespace: List entities or axioms supported as being created by FASTO while using a reused ontology namespace.

entities_with_unexpected_namespace: List entities whose namespace appears inconsistent with their apparent ownership, declaration context, or description in the paper.

detected_issues: This field is an array of issue objects.

For each issue, select exactly one issue_type:

- copied_reuse
- namespace_extension
- namespace_mismatch
- missing_import
- partial_reuse_without_provenance
- local_modification_of_reused_entity
- unexpected_external_namespace
- unclear_ownership
- version_ambiguity
- other

Use other only when none of the listed values applies, and explain the issue in additional_comment. For each issue, select exactly one entity_type:

- class
- object_property
- data_property
- individual
- axiom
- ontology
- unknown

For each issue, select exactly one severity:

- error
- warning
- info

The summary must synthesize only evidence and issues already represented in the corresponding ontology assessment. It must not introduce new conclusions.

Required JSON schema

```
{
  "task": "reuse_namespace_analysis",
  "model": "",
  "input_condition": "Ilm_enhanced",
  "ontology": {
    "name": "FASTO",
    "base_namespace": "",
    "analyzed_reused_ontologies": []
  },
  "reused_ontology_assessments": [
    {
      "reused_ontology": "",
      "expected_original_namespace": "",
      "detected_namespaces": [],

```

```
"import_method": [],
"reuse_scope": "unclear",
"directly_reused_entities": {
  "classes": [],
  "object_properties": [],
  "data_properties": [],
  "individuals": [],
  "axioms": []
},
"local_entities_in_reused_namespace": {
  "classes": [],
  "object_properties": [],
  "data_properties": [],
  "individuals": [],
  "axioms": []
},
"entities_with_unexpected_namespace": {
  "classes": [],
  "object_properties": [],
  "data_properties": [],
  "individuals": [],
  "axioms": []
},
"detected_issues": [
  {
    "id": "RNS1",
    "entity": "",
    "entity_type": "unknown",
    "issue_type": "unclear_ownership",
    "severity": "warning",
    "evidence": "",
    "explanation": "",
    "suggested_fix": "",
    "confidence": 0.0,
    "additional_comment": ""
  }
],
"summary": {
  "main_problem": "",
  "maintenance_risk": "",
  "recommended_action": ""
}
```

```

}
}
],
"global_comments": [
{
"comment": "",
"evidence": "",
"confidence": 0.0
}
]
}

```

Internal quality checks

Before returning the JSON, internally verify that:

- every reused ontology and namespace is supported by evidence;
- every ownership or reuse claim has supporting evidence;
- direct copying was not inferred only from missing owl:imports;
- uncertain cases use appropriate confidence and classification;
- entity-level differences were not incorrectly merged;
- exact or substantively equivalent duplicates were removed;
- every qualifying entity is listed as a separate complete IRI;
- no entity-array element contains prose, a count, or multiple IRIs;
- every exact source-ontology match is included in `directly_reused_entities`;
- every FASTO declaration in a reused namespace was considered for local or unclear ownership;
- every reused entity receiving FASTO-local axioms was considered for `local_modification_of_reused_entity`;
- every grouped issue is supported by complete entity enumeration;
- no placeholder, ellipsis, truncation notice, or omitted-remainder statement occurs;
- all required fields and arrays are present;
- issue IDs are unique, global, and sequential;
- the final output is valid JSON.

Now analyze the supplied evidence and return the downloadable JSON file only.